

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016–17

Zoology**Paper No. ZOB-504 : Biotechniques**

Time: 3 hours

Full Marks: 70

Note: Write your Roll No. at the top immediately on the receipt of this question paper. The figures in the right-hand margin indicate marks. Answer six questions including Question 1 which is compulsory.

Question 1. Answer the following parts:

(a) Select and write the best options in the following:

 $[4 \times \frac{1}{2} = 2]$

(i) In SDS-PAGE, acrylamide determines

- (1) pH of Gel
- (2) pore size
- (3) electrophoretic mobility
- (4) ionic strength

(ii) Resolving power of microscope is dependent upon

- (1) numerical aperture
- (2) eye piece
- (3) magnification
- (4) body tube

(iii) Swelling followed by rupture of cells *in vitro* is due to change in

- (1) pH
- (2) osmolarity
- (3) temperature
- (4) ionic strength

(iv) Substances get separated during centrifugation due to

- (1) centrifugal force
- (2) centripetal force
- (3) gravitational force
- (4) relative centrifugal force

(b) State whether the following statements are True or False. Justify your answer in brief:
[6 × 2 = 12]

- (i) DMSO is not required in culture medium during cryopreservation.
- (ii) The electrochemical potential between two liquids is measured by pH meter.
- (iii) Geiger-Muller counter can detect only beta particles.
- (iv) Serum is not an important requirement for adherent cell culture.
- (v) SEM produces more resolution than TEM.
- (vi) Ethidium bromide is commonly used to detect proteins in PAGE.

(c) Define the following:

[4 × 1 = 4]

- (i) Electrophoresis
- (ii) Antibody
- (iii) Limit of resolution
- (iv) Cell fractionation.

(d) Expand the following:

[4 × $\frac{1}{2}$ = 2]

- (i) TEMED
- (ii) FISH
- (iii) EIA
- (iv) IHC.

Question 2. Describe in detail with labeled diagram the principle and application of phase-contrast microscope. [10]

Question 3. Describe the general composition of cell culture media. Comment upon their uses and significance. [10]

Question 4. What is radioactivity? Describe the measurement and safety aspects of radioactivity. [10]

Question 5. What is microtomy? Enumerate the procedure for the preparation of histological HE stained slides. [10]

Question 6. Write down the principle of centrifugation. Describe the various types of centrifuges used for the separation of biomolecules. [10]

Question 7. Write down the procedure of DNA extraction using mammalian tissue. Describe the principle and application of DNA-gel electrophoresis. [10]

Question 8. Write short notes on the following: [2 × 5 = 10]

(a) Spectrophotometer

(b) Cryopreservation.

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